What survives when a paper leaves the lab: a pre-registered re-test of 22 robot-learning papers on a €1,350 rig

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Abstract

Robot-learning papers are mostly shown in simulation, on costly arms or on data-centre GPUs. We re-tested the central claim of 22 papers on a low-cost arm and an edge GPU, running the cheapest baseline first and fixing a pass mark before the run. Of the 37 claims taken from the papers, 12 reproduce and 18 fail; 8 further rows compare against our own baselines. The failures we see fall into three proposed classes: methods that miss a real-time budget on the edge GPU, zero-shot judges beaten by a trivial signal, and latent planners beaten by the commanded joint positions. We then audited our own pre-registration by commit order, recomputed the early gates with one statistic, and rebuilt every published number from its stored rows. Whether the bar preceded the result is provable for most gates, not for all, and the class assignment awaits a blind second reader; both limits are stated with their counts. Total compute: $45.10.

1. The register

A claim shown on a Franka-class arm, in simulation or on a data-centre GPU does not say whether it holds on the hardware most students and small labs own. The register is one entry per paper: the claim as the paper states it, the pass mark we fixed for our rig, our result and a verdict, with the numbers recomputable from published rows. It rests on three conditions. **Accessible hardware:** an SO-101 arm and a Jetson Orin NX 16 GB, €1,350 for the parts with a stated price (2 smaller parts not yet priced). **Cheapest baseline first:** before a learned model is credited, the trivial signal it must beat (copy-last, the commanded joints, episode length) is run on the same rows. **Bar before the run:** the pass mark is committed before the result; §2 audits how far that holds. Of the 22 papers, 9 were shown in simulation, 7 on rigs above $20k, 3 at $5–20k, 2 below $5k and 1 on an unstated setup.

*Every number with its source file: `github.com/ravediamond/paper-register-reproducibility`.

2. Method, and an audit of our own pre-registration

Each gate of the experiment log has a bar and a result; we find, in git history, the first commit containing each and classify the gate by commit order alone (`git merge-base --is-ancestor`), never from a note or a timestamp. Of the 89 gate rows of the log as of 2026-10-31, **58 are bar-first, 6 same-commit, 0 result-first, 23 no-bar and 2 not-run** (a bar, never a result). The 93 rows of the log today give 61 / 7 / 0 / 23 / 2. "Same-commit" means the bar and the result first appear in the same commit of this repository; it is not evidence that the result came first, nor that it did not. A first pass by text search alone had read 36 bar-first, 31 same-commit, 19 no-bar and 3 unresolved. Tracing each same-commit gate to its own protocol, plan, scorer or specification moved 21 to bar-first and 4 to no-bar (the bar cell itself declares none), and left 6. Those upgrades rest on a bar text found in an earlier commit of a file tied to the gate; the evidence line for each is in the audit table, so a reader can check it. The earlier bar-first rows were not re-audited for a result elsewhere in the history. One gate added on 2026-11-01 (the reward-model progress test) is same-commit by text search, but its protocol (`6241570`) predates the log commit; its result commit was not audited, so it stays same-commit and is an open candidate. **One statistic.** The early gates mixed point estimates and other intervals. We recomputed 10 of them on the episode bootstrap (2,000 resamples, percentile 95 % CI) from stored rows. 3 verdicts flip somewhere inside their interval: the oscillation rate among failed episodes (24.0 %, CI 8.0–40.0 %, bar $\geq$ 10 %, n = 25; pass on the point), a judge's accuracy (94.2 %, CI 89.6–98.9 %, bar $\geq$ 95 %, n = 87; fail) and a judge threshold's false-intervention rate (2.4 %, CI 0.0–7.1 %, bar $\leq$ 5 %, n = 42; pass, but the threshold was fitted to meet that rate, so its interval is descriptive). The register keeps the point verdicts and reports the three as fragile. **Reproducibility of this study.** Every published number was recomputed from its published rows: 29 / 29 write-ups pass. From a fresh clone with no credentials, 6 gates rerun cold and reproduce (5 exactly); 12 are blocked by one unpublished software environment and 11 need a larger GPU, a private checkpoint, the arm or an unstated step. The regis-

ter's citations and paper-side fields were audited against the papers' PDFs and the fixes are merged; of the 38 arXiv ids the published register names, the missing ones in our papers mirror belong to scheduled tests, and one tested paper, which has no arXiv id, was not checked against its PDF.

3. Results

Table 1 lists the register. Of the 22 papers, 4 reproduce, 13 fail, 3 are not proven, 1 was not tested and 1 was withdrawn. The register holds 45 claim rows (42 tested): 17 reproduce, 20 fail, 3 not proven, 4 not tested, 1 withdrawn. 8 of the 45 rows are our own baseline comparisons rather than paper claims (5 of them reproduce, e.g. "the commanded joints beat the world model"); on the 37 paper claims alone, 12 reproduce and 18 fail.

4. Three failure classes

We propose three classes for the 20 failed claim rows. **A, edge latency:** the method may work, but on the edge GPU it misses the real-time budget the claim needs; an adaptive chunking method takes 4,751 ms per decision against a budget of 298 ms. **B, zero-shot judges:** a model used without training on this rig to judge success, failure or progress is no better than a trivial signal; a learned reward model detects failures at AUROC 0.818 where episode length alone reaches 0.938. **C, latent planners beaten by joints:** a learned latent used to plan, rank or track progress does worse than the commanded joints or a copy-last / action-only baseline; a latent world model ranks the executed chunk first in 0.60 of anchors, against a bar of 0.8 and 0.68–0.90 for an action-only baseline. The three examples are illustrative. The assignment of every failed row is pending: the owner labels the 20 rows first and commits the labels, then a blind second reader labels the same form, and the classes are reported as reproducible only if Cohen's $\kappa \geq 0.6$ (the bar was fixed before any reading).

5. Limitations

One rig, one arm, one operator: a failure here is a rig finding, not a refutation of the paper's own setting. The verdicts were labelled by one person, at rest, not always blind to the arm. 6 gates are same-commit and 23 have no bar, so for 29 of 89 gates the order of bar and result is not provable from this repository; the commit-order audit is mechanical, but its 21 upgrades rest on matching a bar text, and the second-reader check of the classes is the only independent check in the study, not yet done. 3 verdicts are fragile under the bootstrap. Only 6 gates were rerun cold. The closest statement of our thesis [5] scores VLA and world-action models on five dimensions, among them reproduction evidence and deployment affordability; its reproduction-evidence score is five audit milestones (configuration mapping, task records, task coverage, training records, seeds) and, in its words, not a probability that a result is correct. We give one pass/fail per claim on a €1,350 rig, with the bar committed before the run.

6. Related work

A standardized real-world benchmark on the same arm reports a failure and recovery taxonomy for VLA policies [2]; a fleet of SO-101 cells evaluates policies at scale [3]. A unified harness decouples VLA inference from simulation benchmarks [1], and a benchmark of demonstration quality studies one method family empirically [4]. These evaluate policies; we re-test the claims behind methods (judges, world models, latency) against bars fixed before the run, and audit our own pre-registration.

References

[1] vla-eval: A Unified Evaluation Harness for Vision-Language-Action Models, 2026. arXiv:2603.13966.

[2] Benchmarking Vision-Language-Action Models on SO-101: Failure and Recovery Analysis, 2026. arXiv:2606.08881.

[3] ArmnetBench v0.1: Parallel Real-World Evaluation of Manipulation Policies on a Low-Cost Arm Farm, 2026. arXiv:2607.24481.

[4] What Matters in Quality-Aware VLA Pretraining: A Benchmark and Empirical Study, 2026. arXiv:2610.18869.

[5] Xin Ma et al. RoboSeed: Multidimensional Evaluation of Vision-Language-Action and World-Action Models, 2026. arXiv:2610.40026.

paper	their rig	claim tested	bar	our result	verdict	class
LeWorldModel	sim	A small JEPA world model learns the arm's dynamics well enough to plan with it.	Dynamics: beat copy-last, every corruption ≥ 1.3 at $K = 30$. Planning: rank the executed chunk first among 8 in ≥ 0.8 of anchors.	dynamics 0.31 vs 0.89 copy-last (reproduces); planning top-1 0.60 vs 0.68–0.90 for an action-only baseline (fail)	fail	pending
DINO-WM	sim	A predictor on frozen DINOv2 patches plans better than one on a learned latent.	G1b top-1 ≥ 0.8 at some target distance.	arm-pose R^2 0.92 (vs 0.57 for W1); G1b top-1 0.55–0.61; no-model baseline 0.68–0.90	fail	pending
RoboJEPA	>$20k	Pretrained RoboJEPA predicts our latents, uses the actions, and runs on the robot's computer.	9-step L1 ≥ 15 % below copy-last (300M) and ≥ 5 % below shuffled actions; 2-view encoder ≤ 200 ms on the Orin.	11.8 % below copy-last (300M; 1B 15.8 % on 30 clips); 9–10 % below shuffled; encoder 526 ms on the Orin NX	fail	pending
H-JEPA	sim	A slow second level makes latent distance track task progress.	Progress monotonicity Spearman $\rho \leq -0.80$; false stalls ≤ 10 %.	level-2 ρ -0.73; false stalls 52 %	fail	pending
D-JEPA	<$5k	A decision-aligned ranking operator on a frozen world model picks better action chunks.	G1b regret ≤ 0.10 and $\leq$ the action-only baseline + 0.05.	regret 0.263 (W1 alone 0.263; action-only -0.007)	not tested	—
ContactGuard	$5–20k	An event-anchored probe on imagined latents predicts grasp failure better than the current latent.	AUROC ≥ 0.85 and ≥ 0.05 above the current latent.	0.70 imagined vs 0.82 current	fail	pending
FAIL-Detect and SAFE	>$20k	A conformal band on a learned failure score keeps false alarms at α while catching failures.	Successes flagged $\leq 1/20$; failures flagged ≥ 50 %.	22.7 % of successes flagged at $\alpha =$ 5 %; 46 % of failures caught	fail	pending
ReAP	$5–20k	A risk head on an action-conditioned latent warns before failures.	$\geq 5/8$ failures caught at $\leq 1/20$ false interventions; action-conditioned iff true $\geq$ shuffled + 0.05.	v1 AUROC 0.93 came from time and length; v2 0.789 vs 0.708 shuffled, 3/8 at 5/42	fail	pending
VLA-Corrector	sim	A verifier on observable history flags a failing rollout early enough to correct it.	$\geq 5/8$ failures at $\leq 1/20$ false interventions.	the paper's verifier family on our rollouts (a GRU on pooled RGB statistics and actions): 2/8 caught at 3/42 false interventions, AUROC 0.59; cosine drift of the W1 latent 0/8, AUROC 0.30	fail	pending
Robometer	>$20k	A 4B reward model judges SO-101 rollouts zero-shot.	Pooled AUROC ≥ 0.85 and within-policy ≥ 0.8 on ≥ 5 of 7 policies.	pooled AUROC 0.818 on 2099 ArmnetBench rollouts; episode length alone 0.938	fail	pending
SmolVLA	<$5k	A 450M VLA succeeds on an SO-101 task and can run on low-cost hardware.	10-trial operator-judged gate (no numeric pass mark registered); chunk inference within the async budget (667 ms).	9/10 on bowls; 974 ms in PyTorch, 238 ms with vla.cpp on the Orin NX	reproduces	—

Table 1: The register: one row per paper, generated from the register file (claim, bar and result as recorded, including our internal model and gate names). "Class" is the failure class of §4, pending the labels of §4.

paper	their rig	claim tested	bar	our result	verdict	class
TurboVLA	sim	Our claim that the architecture made the policy ignore its instruction.	Task swap $\geq 5°$ and per-layout $\geq 75\,\% \rightarrow$ our port was the bug; $< 1°$ with the control $\geq 0.095 \rightarrow$ the data.	upstream's own trainer on our data: task swap 0.020°, per-layout 38.9 %; control, their released LIBERO checkpoint run here: 0.105 ($\approx 5.5°$ of pan)	withdrawn	—
VLA-JEPA	sim	VLA-JEPA trains cheaply on SO-101 data and runs on the robot.	10-trial operator-judged gate (no numeric pass mark registered); for reranking (H0): oracle / mean error $\leq$ 0.7 and N = 8 $\leq 1.5\times$ the latency.	9/10 for $9.7; 355 ms per chunk on the Orin NX; reranking no-go (0.76, 1.84×)	reproduces	—
REKI	sim	Policies without memory oscillate between poses at a measurable rate.	$\geq 10\,\%$ of failed episodes oscillate (their criterion).	20.3 % of all episodes, 24.0 % of failures; REKI trained on my data: 19/20 + 19/20 on the arm, oscillates in 1/40	reproduces	—
Real-Time Chunking	unstated	Conditioning a new chunk on the executing one removes the jump at chunk boundaries.	Best gradient-free method cuts the median jump $\geq 50\,\%$ against naive async, drift $\leq +10\,\%$, within 667 ms.	prefix overwrite cuts the jump 71–73 % against naive async at +0.4 ms; tuned LeRobot identity mode is close	not proven	—
HWM (hierarchical latent world models)	>$20k	A high-level world model over learned macro-actions beats the low-level model beyond about 1 s (their Fig. 6 crossover).	Crossover in [1.0, 2.0] s; at 2 s high-level error $\leq 0.95 \times$ low-level; learned macro-actions $\geq$ net joint displacement on cosine.	no crossover: W1 0.382 $\pm$ 0.005 $\rightarrow$ 0.563 $\pm$ 0.007, high level 0.769 $\pm$ 0.009 $\rightarrow$ 0.759 $\pm$ 0.009 (0.73–2 s); cosine 0.608 $\pm$ 0.007 vs 0.551 $\pm$ 0.006	not proven	—
LeJEPA identifiability	sim	A SIGReg JEPA trained on goal-directed data recovers the true latent state only partially and unevenly.	W1's mean R^2 from latent to joints < 0.80; anisotropy (max $-$ min per-joint R^2) ≥ 0.20; the higher-R^2 model has the higher G1b top-1.	W1 mean R^2 0.565, anisotropy 0.639 (shoulder lift 0.91, wrist flex 0.27); frozen DINOv2-S 0.867 / 0.301	reproduces	—
SURVEYOR	sim	A threshold on the imagined-vs-reached latent distance accepts good plans and rejects bad ones (τ = 0.20).	τ below the 10th percentile of non-equivalent pairs and ≤ 0.40; as a monitor $\geq 5/8$ failures at $\leq 2/42$; ≥ 0.05 AUROC over a shuffled drafter.	$\tau = 1.249$ on W1 latents of 69 real rollouts; as a monitor 3/8 at 6/42 (S = 30), AUROC 0.756 vs shuffled 0.560	fail	pending
Adaptive Action Chunking (AAC)	>$20k	Chunk length chosen from the entropy of N sampled chunks follows the task's phases (long in transport, short at the grasp), lowers the boundary jump, and costs little with batched sampling.	median h* in transport $\geq$ grasp window + 2; jump with AAC + overwrite $\leq$ fixed + overwrite at d = 8; an N = 20 decision ≤ 298 ms on the Orin.	h* 2 in transport and 2 at the grasp (86 % of decisions at the floor); jump 0.973 vs 0.940; N = 20 decision 4,751 ms (sequential) on the Orin	fail	pending
PACT-VLA	$5–20k	Picking among N sampled chunks by their flow residual under the pre-commit observation fits what happens next better than a random or stale pick, with no added control-step delay.	PACT beats a random pick and the stale-observation residual at d = 8 (CIs < 0); its delay on the Orin $\leq$ the naive delay; at that delay it still beats a single sample.	offline $-0.27°$ vs random and vs the stale residual (both CIs < 0); on the Orin 1,200 ms per cycle, 37 control ticks vs 8	fail	pending
Horizon × depth (Euler steps vs execution horizon)	>$20k	With long open-loop commitment, cutting a flow-matching VLA's Euler steps from 10 to 2 contracts the action cloud around an unchanged centroid, saves most of the inference time and avoids edge-GPU thermal throttling.	median R(2)/R(10) ≤ 0.70 with the centroid within 0.05 relative and 3°; N = 2 latency $\leq 0.85 \times$ N = 10 on the Orin, N = 10 within 5 % of 238 ms.	R(2)/R(10) 0.78 (CI 0.73–0.81), centroid 0.018 and 2.35°; 196.1 vs 237.5 ms per chunk (0.83, vision 151 ms); no throttling at N = 10 on the Orin, at MAXN_SUPER or 25 W	not proven	—
REMORA	>$20k	Reading a progress distribution out at the supported peak nearest the previous estimate steadies the estimate and lets it fall during real setbacks, with no worse detection of the stages.	Against per-frame argmax: premature not higher and missed $\leq$ +2 points (P1); flips halved, CI upper ≤ 0.7 (P2); a decrease inside $\geq 50\,\%$ of the reversal windows with $\leq 20\,\%$ false regressions (P3).	premature 0.22 vs 0.25, missed equal (P1 pass); flips 0.70 × argmax, CI 0.48–0.94 (P2 fail); setback recall 2/7, false 0.52 (P3 fail); MAE 0.184 vs median 0.218	fail	pending

Table 1, continued.